RISK OPERATIONS DASHBOARD

Project Report & Technical Documentation

v1.0

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1. Project Overview

The Risk Operations Dashboard is a full-stack internal web application built for financial risk operations teams. It provides real-time merchant risk monitoring, bulk compliance operations, feature management, and a full compliance audit trail — all behind role-based access control.

The application was developed iteratively, starting from a basic merchant list and evolving through several sessions into a production-ready dashboard with login, role gating, bulk operations, and resilient API handling.

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2. Technology Stack

Layer	Technology
Backend	Java 17, Spring Boot 3.2.0
Database	H2 (in-memory, dev profile)
ORM	Spring Data JPA / Hibernate 6.3
Security	Spring Security — HTTP Basic Auth
Frontend	Vanilla HTML / CSS / JavaScript (single-file SPA)
Build Tool	Apache Maven 3

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3. Project Structure

```
risk-ops-dashboard/
├── pom.xml
├── src/
│   ├── main/
│   │   ├── java/com/fintech/riskops/
│   │   │   ├── RiskOpsDashboardApplication.java
│   │   │   ├── config/        # Security, CORS
│   │   │   ├── controller/    # REST controllers
│   │   │   ├── dto/           # Request/Response DTOs
│   │   │   ├── model/         # JPA entities
│   │   │   ├── repository/    # Spring Data repos
│   │   │   └── service/       # Business logic
│   │   └── resources/
│   │       ├── application.yml
│   │       ├── application-h2.yml
│   │       └── static/
│   │           └── index.html  <-- EDIT THIS FILE
│   └── test/...
```

Important: Always edit src/main/resources/static/index.html. The target/ directory is Maven's build output — it is auto-regenerated on every build and any manual changes to it will be overwritten.

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4. User Roles & Access Control

Role	Username	Password	Access Level
Analyst	analyst	analyst123	Read-only: view, search, evaluate risk
Risk Ops	riskops	riskops123	Operations: create, bulk freeze/unfreeze, feature toggles
Admin	admin	admin123	Full access: all operations + CSV export

Role-Based UI Restrictions

Feature	Analyst	Risk Ops	Admin
View merchants / audit log	Yes	Yes	Yes
Run risk evaluation	Yes	Yes	Yes
Create merchant	No	Yes	Yes
Bulk freeze / unfreeze	No	Yes	Yes
Enable / disable features	No	Yes	Yes
Export CSV	No	No	Yes

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5. API Endpoints

Base URL: http://localhost:8080/api/v1 | Auth: HTTP Basic

Merchants

Method	Endpoint	Description
GET	/merchants?size=50	List all merchants
POST	/merchants	Create a new merchant
GET	/merchants/dashboard/metrics	Dashboard summary metrics
POST	/merchants/bulk/freeze	Bulk freeze selected merchants
POST	/merchants/bulk/unfreeze	Bulk unfreeze selected merchants

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POST	/merchants/bulk/feature/enable	Enable a feature for multiple merchants
POST	/merchants/bulk/feature/disable	Disable a feature for multiple merchants

Risk Evaluation

Method	Endpoint	Description
POST	/risk/evaluate/{merchantId}	Run risk evaluation for a merchant

Audit Log

Method	Endpoint	Description
GET	/audit?size=50	Get all audit log entries
GET	/audit?actionType={type}&size=50	Filter by action type
GET	/audit/export/csv	Export as CSV (ADMIN only)

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6. Frontend Features

The entire frontend is a single HTML file. It functions as a single-page application with no external framework dependencies.

Login Screen

- Three role cards: Analyst, Risk Ops, Admin
- Clicking a role opens a password modal — no immediate login
- Password validated client-side before granting access
- Session persisted in sessionStorage across browser refreshes
- Wrong password shows inline error and clears input field

Dashboard Page

- Live metrics: flagged today, frozen accounts, high-risk count, pending reviews
- Risk distribution grid (LOW / MEDIUM / HIGH / FROZEN)
- Auto-refreshes every 30 seconds

Merchants Page

- Full merchant table with name search and risk status filter

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- Risk score progress bar, colour-coded by severity
- Feature pills showing enabled/disabled features (PAY / WIT / API)
- Per-merchant risk evaluation button
- Checkbox multi-select with sticky bulk action bar
- Bulk action buttons hidden entirely for ANALYST role

Audit Log Page

- Compliance timeline with colour-coded event dots
- Filter by action type dropdown
- Shows: action type, entity, timestamp, operator, old/new value, description
- CSV export button visible to ADMIN only

Modals

- Password Modal — validates role credentials, shows error on failure
- Bulk Freeze Modal — requires reason field (compliance requirement)
- Bulk Unfreeze Modal — choose target status and provide reason
- Add Merchant Modal — name, email, business type, risk status, feature checkboxes

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7. Key Technical Decisions & Fixes

7.1 API Timeout with AbortController

All API calls use a 5-second timeout via AbortController. If the backend does not respond, the request is aborted and the frontend falls back to built-in mock data automatically. This prevents the blank/buffering screen issue caused by hanging fetch() requests.

```
var controller = new AbortController();
var timeout = setTimeout(() => controller.abort(), 5000);
opts.signal = controller.signal;
var res = await fetch(API_BASE + path, opts);
clearTimeout(timeout);
```

7.2 Interval Guard on Login

Each login previously created a new setInterval() without clearing the old one, causing multiple background timers to stack. Fixed by always clearing the existing interval before creating a new one.

```
if (refreshInterval) clearInterval(refreshInterval);
refreshInterval = setInterval(loadMetrics, 30000);
```

7.3 Dynamic API Base URL

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The API base URL is derived from window.location.origin so the frontend works on any hostname or port without hardcoded values.

```
const API_BASE = window.location.origin + '/api/v1';
```

7.4 Role-Based UI Visibility

After login, the showAppScreen() function checks CURRENT_ROLE and hides or shows buttons accordingly. This prevents ANALYST users from seeing or accidentally triggering operations they are not permitted to perform.

7.5 Password Gate on Login

Previously, clicking a role card logged the user in immediately. A password modal was added that intercepts the click, prompts for credentials, validates against the ROLE_PASSWORDS map, and only then calls loginAs(). Wrong passwords show an inline error.

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8. Database

The application uses an in-memory H2 database under the active Spring profile h2.

Property	Value
H2 Console URL	http://localhost:8080/h2-console
JDBC URL	jdbc:h2:mem:riskopsdb
Username	SA
Password	(empty)
Data persistence	In-memory — resets on each restart

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9. Common Issues & Fixes

Problem	Root Cause	Fix
Blank/buffering page	Browser cache serving old file	Ctrl+Shift+R or open Incognito
Chrome shows Google	URL typed without http://	Use http://localhost:8080
Changes not reflected	Edited target/ instead of src/	Edit src/main/resources/static/index.html only
Old page after edits	Stale Maven build	Run mvn clean then mvn spring-boot:run

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Multiple timers stacking	setInterval not cleared on re-login	Fixed — interval cleared before each new one
API calls hanging	fetch() waiting indefinitely	Fixed — 5s AbortController timeout added
Analyst sees bulk buttons	No role check on UI render	Fixed — buttons hidden based on CURRENT_ROLE
Login without password	No password gate on cards	Fixed — password modal required before login

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10. Running the Application

Development

```
cd E:\risk-ops-dashboard
mvn clean
mvn spring-boot:run
```

Then open http://localhost:8080 in your browser.

Development Workflow

- Edit: src/main/resources/static/index.html
- Stop server: Ctrl+C in terminal
- Clean: mvn clean
- Start: mvn spring-boot:run
- Reload: Ctrl+Shift+R in browser

Production Build

```
mvn clean package
java -jar target/risk-ops-dashboard-1.0.0.jar
```

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11. Mock Data

The frontend includes 10 built-in mock merchants and 7 audit log entries. These are used automatically as a fallback when the backend API is unreachable or returns an error, ensuring the UI remains functional for demos and development without a running backend.

Mock Merchant	Status	Risk Score	Business Type
GreenLeaf Payments	LOW	5	E-Commerce
Sunrise Tech Ltd	LOW	12	SaaS

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FastCash Express	MEDIUM	52	Money Transfer
Meridian Trading Co	MEDIUM	48	Trading
VelocityPay Corp	HIGH	78	Payments
Global Remit Hub	HIGH	82	Remittance
Shadow Commerce LLC	FROZEN	94	E-Commerce
QuickDraw Finance	FROZEN	91	Lending
ClearPath Merchants	LOW	18	Retail
Horizon Finserv	MEDIUM	61	Financial Services

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